Day - 18
Functions in Python

SKILL COURSE

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What is a Function?

✓ Layman Explanation

A **function** is like a machine or a mini-program:

You give input
Machine processes it
Machine gives output

Example in real life:

Mixer → Input: vegetables → Output: paste

ATM → Input: card, PIN → Output: cash

In Python:

A function takes **input**, performs **action**, and gives an **output**.

✓ Technical Definition

A function is a reusable block of code defined using def keyword that performs a specific task and optionally returns a value using return.

Why Functions Are Important for Data Analysts?

- Reuse same logic multiple times
- Clean data consistently
- Create custom cleaning functions
- Avoid writing repeated code
- Improve project structure
- Make code readable & modular
- Create reusable transformations
- Essential for Pandas & real projects

Function Practice

- **Basic Syntax of a Function**

```
def function_name():
    # code
```

- **Function With No Parameters**

```
def greet():
    print("Hello, Welcome!")
```

- **Calling a Function**

```
greet()
```

Function With Parameters (Arguments)

```
def welcome(name):
    print("Welcome,", name)

welcome("Satish")
```

Function With Return Values

```
def add(a, b):
    return a + b

result = add(5, 10)
print(result)
```

Functions vs Print (Very Important)

✓ print() → shows output only

✓ return → gives data back so you can use it again

DEFAULT Parameter

```
def greet(name="Guest"):
    print("Hello", name)

greet()
greet("Rohit")
```

Multiple Parameters

```
def full_name(first, last):
    return first + " " + last
```

Practical data analyst function examples

Example 1: Clean Text

```
def clean_text(value):
return value.strip().title()
print(clean_text("   mUMbai   "))
```

Example 2: Fix City Spellings

```
def fix_city(city):
city = city.lower().strip()
city = city.replace("mombai", "mumbai")
city = city.replace("kolkatta", "kolkata")
return city.title()
```

Example

- **Example 3: Extract Year From Product Code**

```
def get_year(code):
    return code[-4:]
```

- **Example 4: Email Validation**

```
def is_valid_email(email):
    return "@" in email and "." in email
```

Function Returning Multiple Values (Tuple)

```
def stats(nums):
    return min(nums), max(nums), sum(nums)/len(nums)

print(stats([10, 20, 50, 80]))
```

Function With Loop Inside

```
def count_vowels(text):
    vowels = "aeiou"
    count = 0

    for ch in text.lower():
        if ch in vowels:
            count += 1

    return count
```

Clean List Using Function

```
def clean_list(values):
    cleaned = []
    for v in values:
        cleaned.append(v.strip().title())
    return cleaned

print(clean_list([" pUne ", "  mUmbai", "DELHI  "]))
```